CS 340 README

About the Project/Project Title

CRUD Python Module for MongoDB (AAC AnimalShelter Class)

This project is a Python module that provides full CRUD (Create, Read, Update, Delete) functionality for a MongoDB database. The module connects to the Austin Animal Center (AAC) database and allows users to insert, retrieve, update, and delete animal records using Python. The functionality is implemented using an object-oriented approach through the AnimalShelter class, making the code reusable and easy to integrate into larger applications such as dashboards.

This project was extended in Project Two by integrating the CRUD Python module into an interactive dashboard application. The dashboard allows users to filter, visualize, and analyze animal data from the MongoDB database in real time using a data table, chart, and geolocation map.

Motivation

The motivation behind this project is to simplify interactions with a MongoDB database by using Python instead of manually executing database commands. This module allows developers to programmatically perform database operations, which is essential for building applications such as dashboards that require automated data access and manipulation.

In addition to simplifying database operations, this project demonstrates how backend data systems can be connected to a frontend interface. The dashboard improves usability by allowing users to interact with the data visually, making it easier to identify animals that meet specific rescue criteria.

Getting Started

To run this project locally, follow these steps:

Note: This template has been adapted from the following sample templates: Make a README, Best README Template, and A Beginners Guide to Writing a Kickass README.

1. Import the AAC dataset into MongoDB using the mongoimport tool.
2. Create a MongoDB user account (aacuser) with read and write access to the aac database.
3. Open Codio and navigate to the code_files directory in JupyterLab.
4. Ensure that the CRUD_Python_Module.py file is present.
5. Open the ProjectOneTestScript.ipynb notebook.
6. Import the AnimalShelter class and instantiate it using your database credentials.
7. For Project Two, open the ProjectTwoDashboard.ipynb file in JupyterLab and run all cells to launch the dashboard. Ensure that the MongoDB credentials match those used in the CRUD module.

During development, one challenge was ensuring that the database connection worked properly with authentication. This was resolved by correctly configuring the MongoDB connection string and specifying the authentication database. Another challenge was handling query results, which was solved by converting the MongoDB cursor into a list.

Installation

The following tools and technologies were used:

- Python: Used to develop the CRUD module
- MongoDB: NoSQL database used to store AAC data
- PyMongo: Official MongoDB driver for Python used to connect and interact with the database
- Jupyter Notebook: Used to test and execute Python code
- Codio: Cloud-based development environment
- Additional tools used in Project Two include Dash (Plotly) for building the interactive dashboard and Dash Leaflet for implementing the geolocation map. These tools were selected because they allow seamless integration of data visualization and interactivity within a Python environment.

PyMongo was selected because it is the official and most reliable library for integrating Python with MongoDB.

Usage

The module allows users to perform full CRUD operations on the MongoDB database.

Example of inserting a document:

```
shelter.create({
    "name": "Test Animal",
    "breed": "Test Breed",
    "outcome_type": "Adoption"
})
```

Example of retrieving documents:

```
results = shelter.read({"breed": "Test Breed"})
```

Example of updating documents:

```
shelter.update(
    {"breed": "Test Breed"},
    {"outcome_type": "Updated Outcome"}
)
```

Example of deleting documents:

```
shelter.delete({"breed": "Test Breed"})
```

Note: This template has been adapted from the following sample templates: Make a README, Best README Template, and A Beginners Guide to Writing a Kickass README.

The dashboard extends this functionality by allowing users to visually interact with the data. Users can filter animals based on rescue categories such as water, wilderness, and disaster/tracking. When a filter is selected, the data table updates with matching records, the chart updates to reflect the filtered dataset, and the map displays the selected animal's location.

Code Example

from CRUD_Python_Module import AnimalShelter shelter = AnimalShelter() data = {"name": "Test Animal", "breed": "Test Breed"} shelter.create(data) results = shelter.read({"breed": "Test Breed"}) for r in results: print(r)

CRUD Functionality

This module implements full CRUD functionality for interacting with the MongoDB database:

- Create: Inserts a new document into the collection and returns True if the operation is successful, otherwise False.
- Read: Retrieves documents based on a query and returns the results as a list.
- Update: Modifies existing documents that match a query and returns the number of documents updated.
- Delete: Removes documents that match a query and returns the number of documents deleted.

Tests

The module was tested using a Jupyter Notebook.

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- The create() function was tested by inserting a sample document and verifying it returned True
- The read() function was tested by retrieving inserted documents
- The update() function was tested by modifying existing records and confirming changes
- The delete() function was tested by removing records and verifying they no longer exist
- Additional testing was performed using invalid and edge-case inputs. The create() function was tested with empty data and correctly returned False. The read() function was tested with queries that returned no matching results and returned an empty list. The update() and delete() functions were tested with queries that matched no records and correctly returned zero modified or deleted documents. These tests confirm that the module behaves reliably under both normal and edge-case conditions.

Error Handling and Edge Case Testing

- The CRUD module was tested with both valid and invalid inputs to ensure reliability. The create() function handles empty or invalid data without causing system failure. The read() function consistently returns a list, even when no records are found.
- The dashboard was also tested under conditions where filters returned no results. In these cases, the application continued to function without crashing, demonstrating stability and proper error handling.

Screenshots

Figure 1: MongoDB import execution (Module Three)

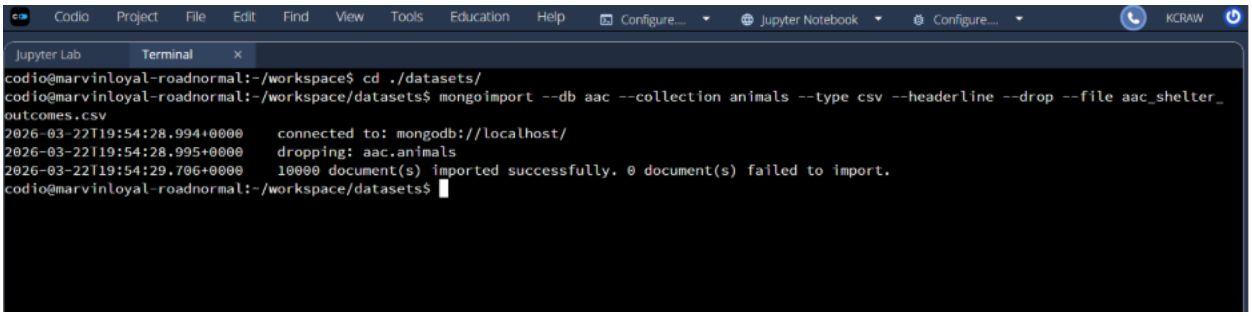
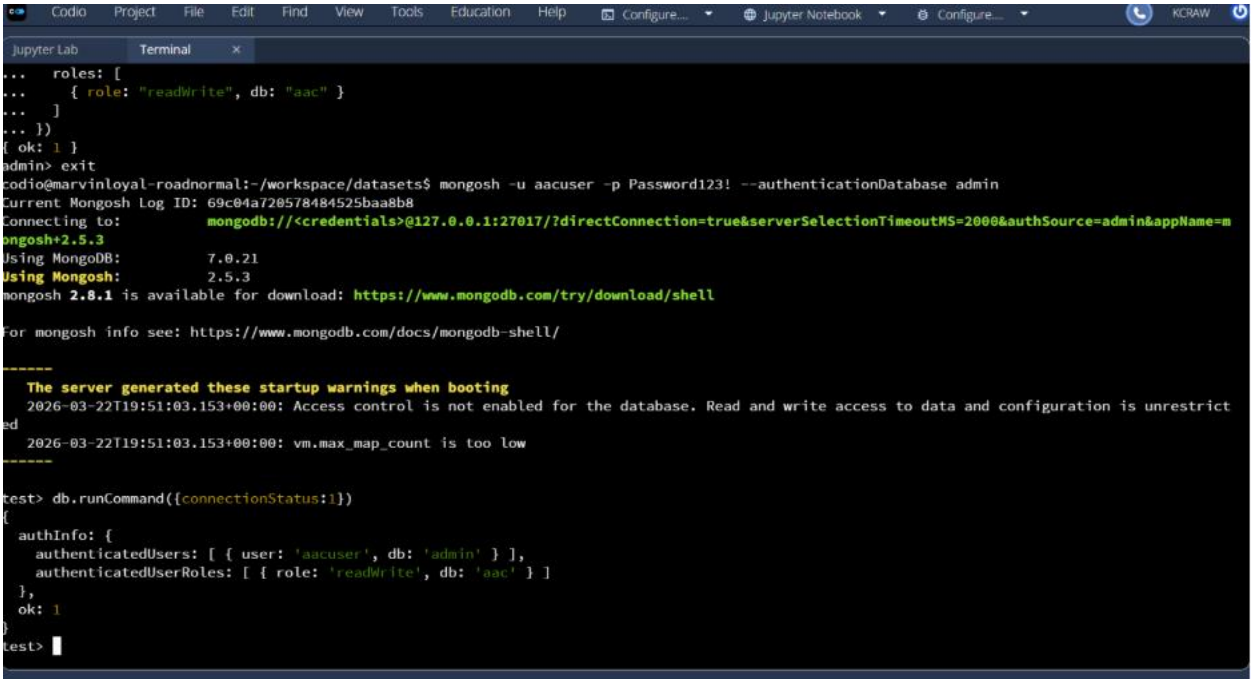


Figure 2: User authentication login (Module Three)



Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

Figure 3: Create function execution showing successful insertion

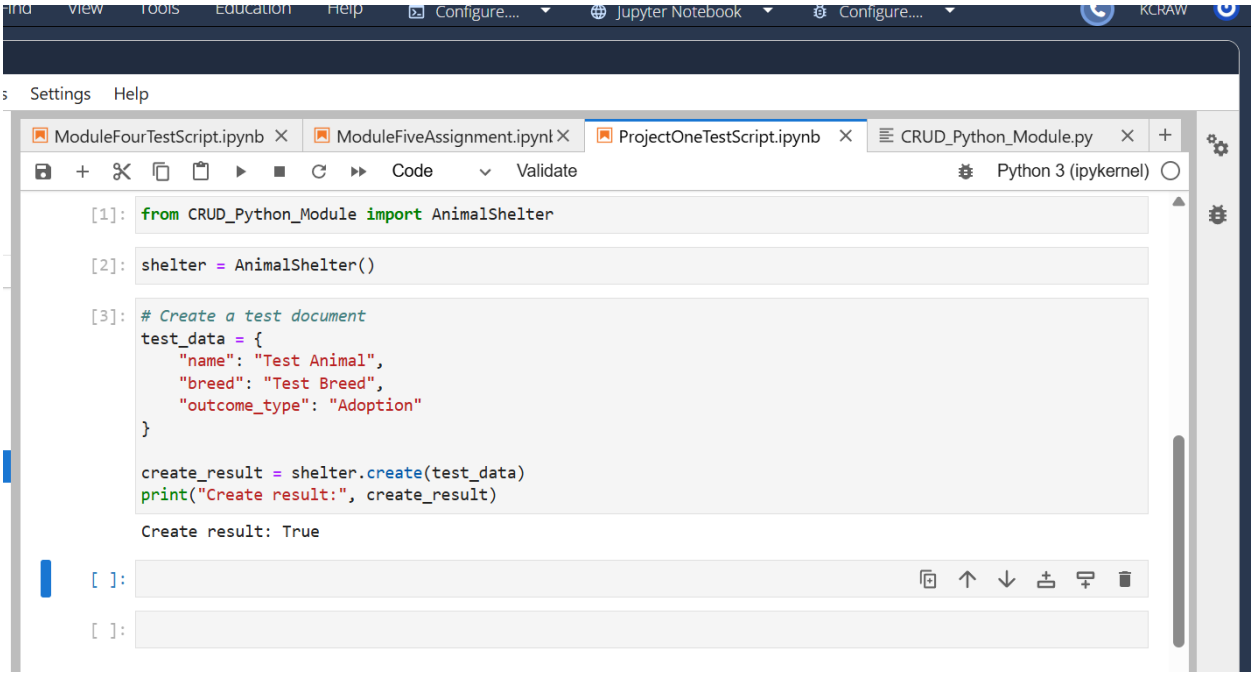
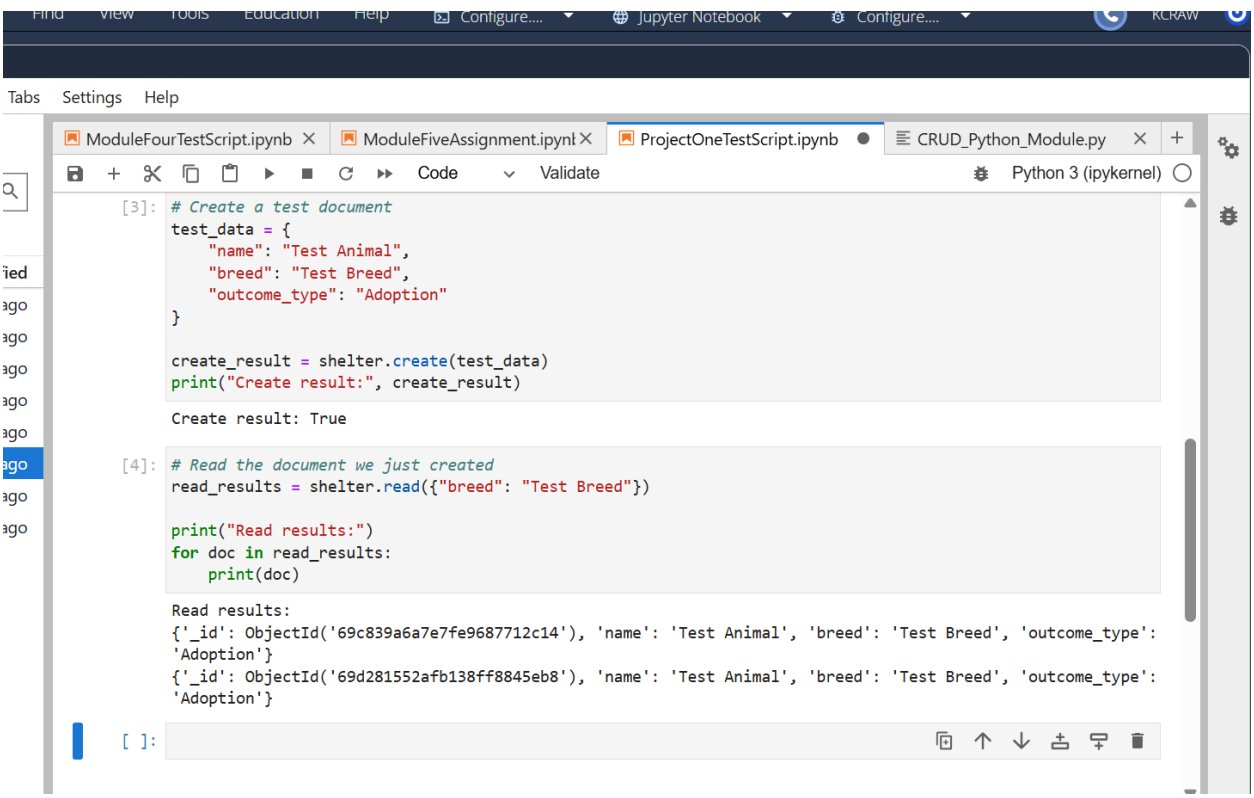
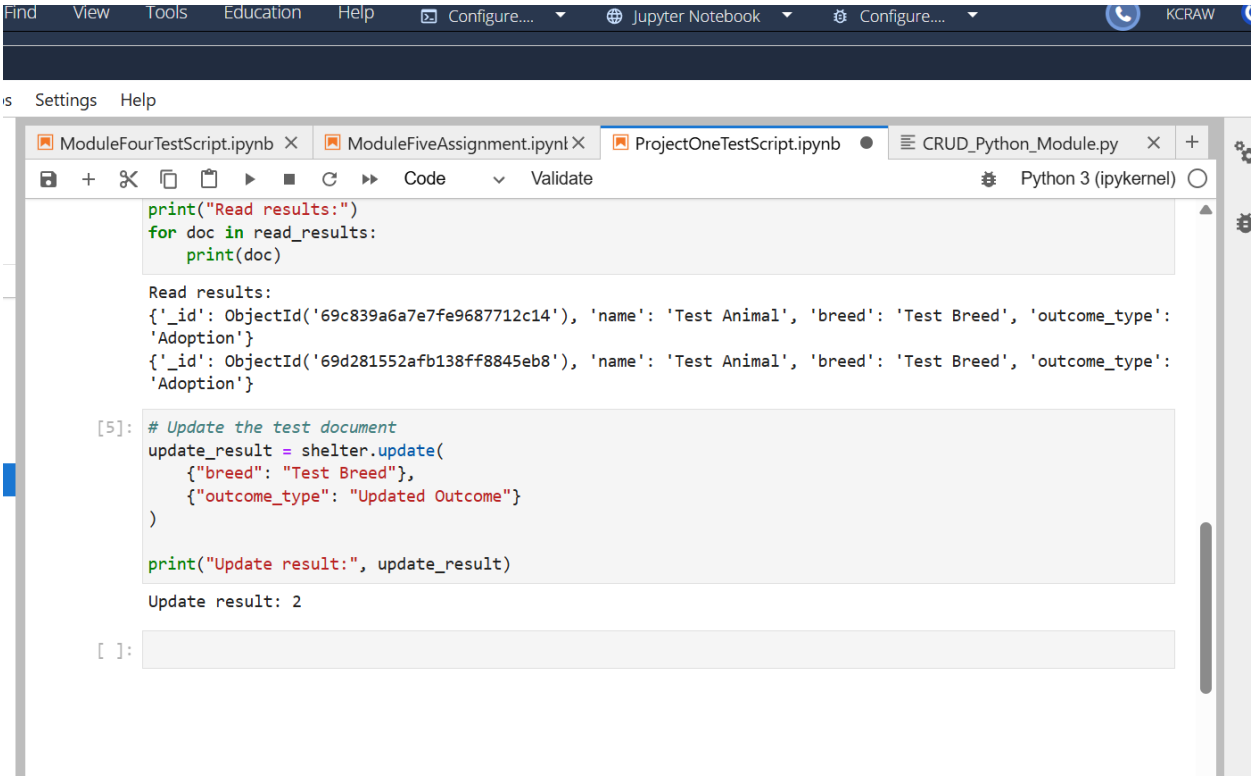


Figure 4: Read function execution showing returned document(s)



Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

Figure 5: Update function execution showing number of modified records



Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

Figure 6: Verification of updated document(s)

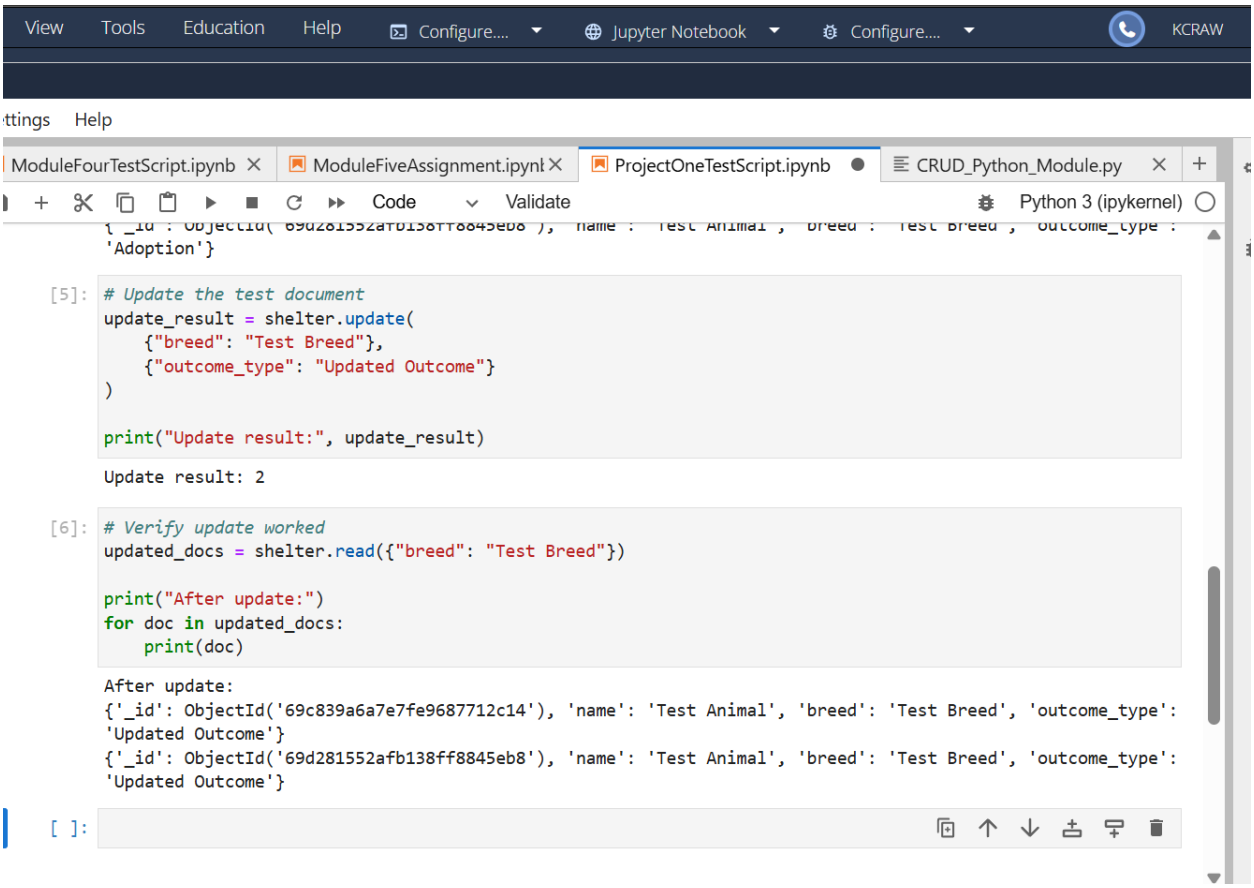


Figure 7: Delete function execution showing number of deleted records

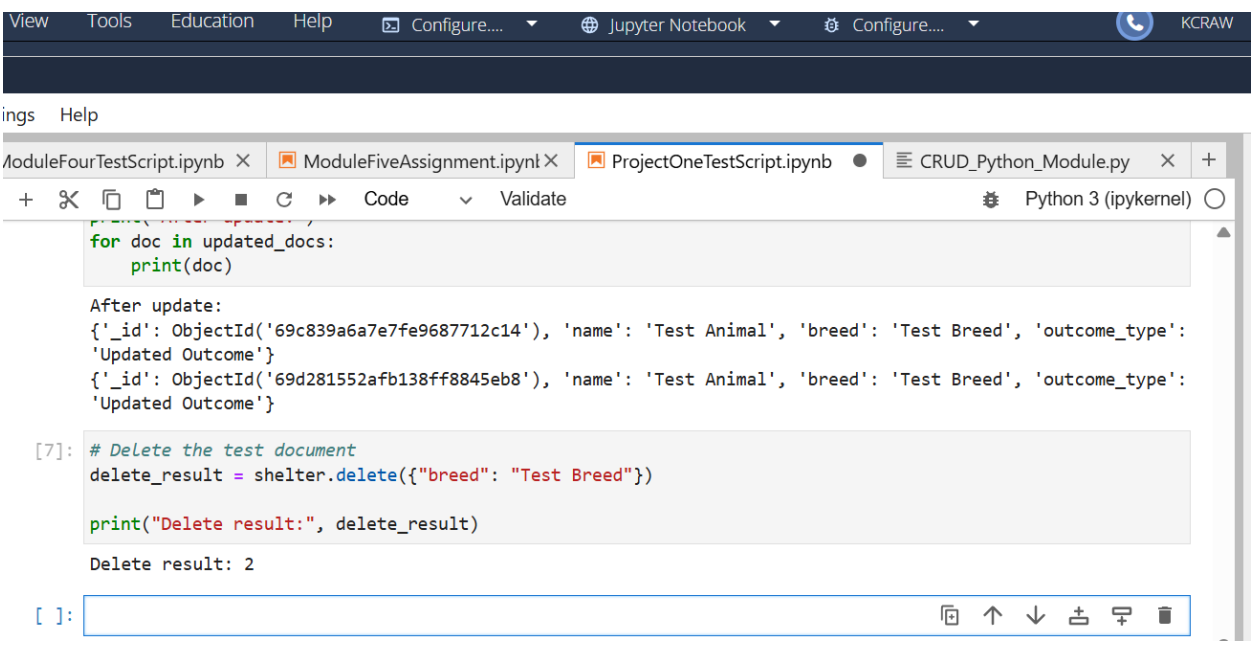


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Figure 8: Verification of deletion (empty result set)

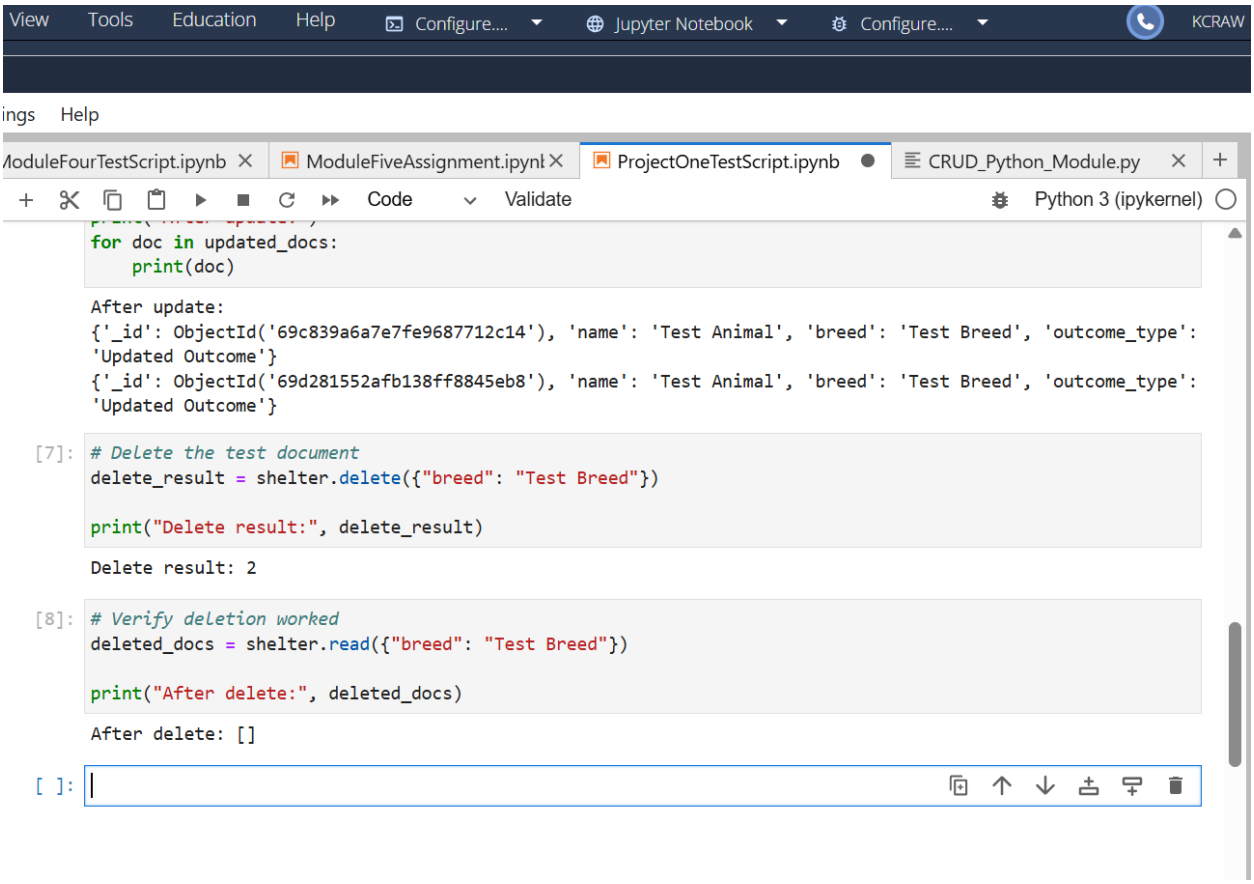

Figure 9: Dashboard initial view (unfiltered data)

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

Southern New Hampshire University

Kenya Craw Dashboard

GRAZIOSO SALVARE

○ Water Rescue ○ Wilderness Rescue ○ Disaster/Tracking ◉ Reset

rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outc
filter										
1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		
3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank	
4	7 months	A733653	Cat	Siamese Mix	Seal Point	2016-01-25	2016-08-27 18:11:00	2016-08-27T18:11:00	Kitty	
5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke	
2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06T10:49:00		
7	2 years	A673830	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15:00	2014-03-19T15:15:00		
8	1 year	A736551	Dog	Labrador Retriever/Australian Cattle Dog	Black	2015-10-12	2016-11-27 18:00:00	2016-11-27T18:00:00		
9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing	
6	5 years	A696004	Dog	Cardigan Welsh Corgi Mix	Sable/White	2010-01-27	2015-01-28 10:39:00	2015-01-28T10:39:00	Lucy	
10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor	

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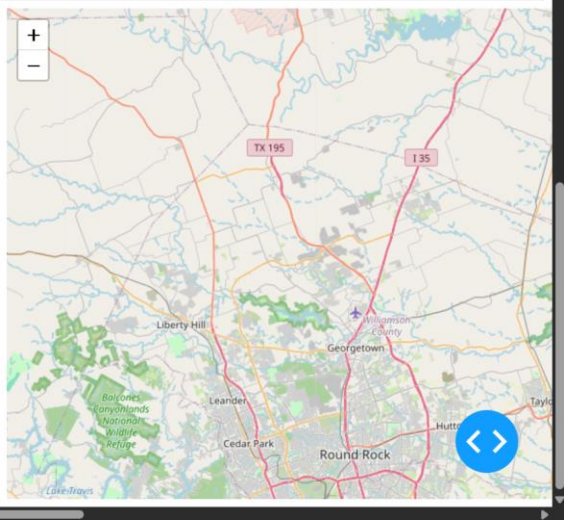


Figure 10: Water Rescue filter applied

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

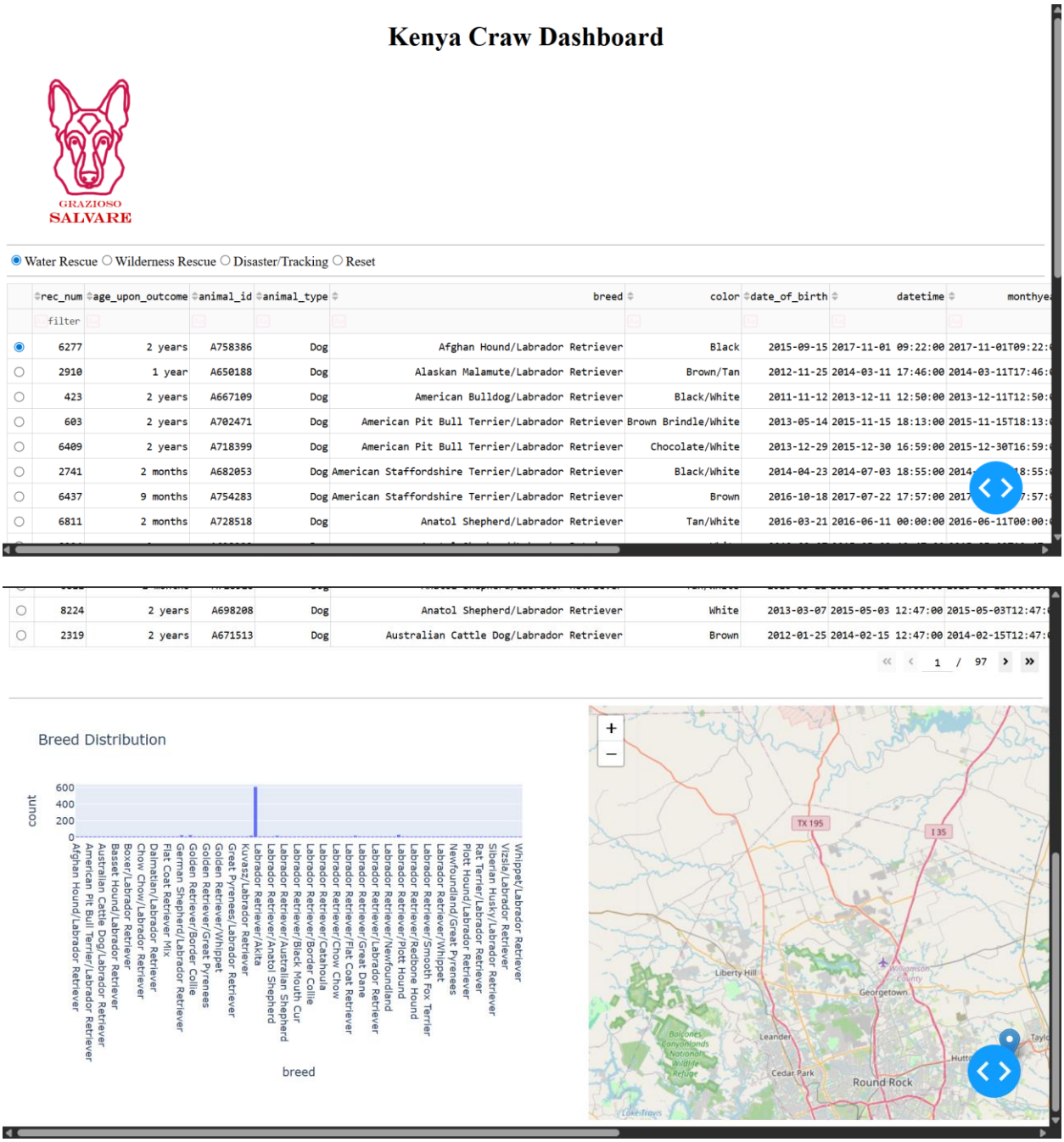


Figure 11: Wilderness Rescue filter applied

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

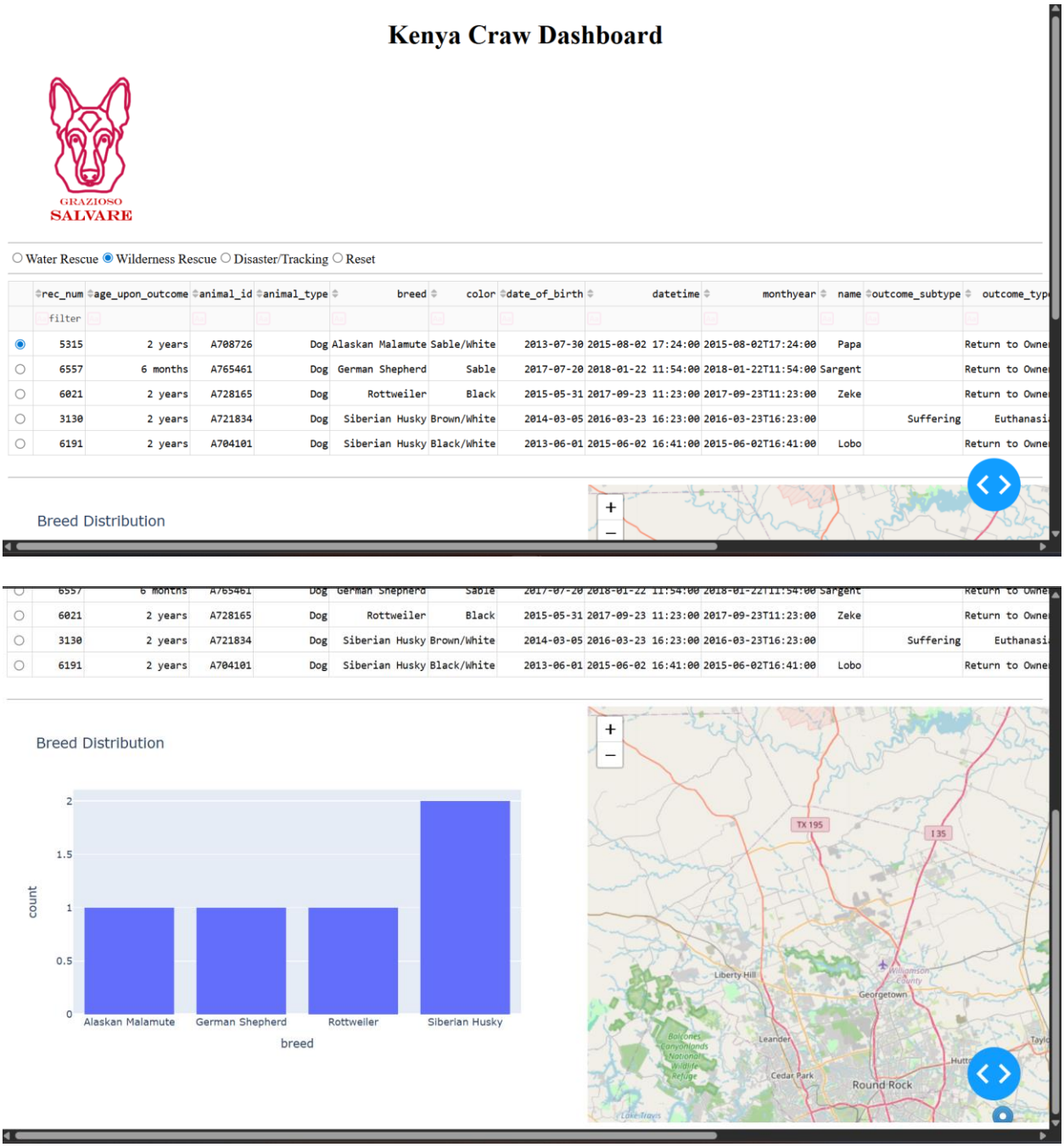


Figure 12: Disaster/Tracking filter applied

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

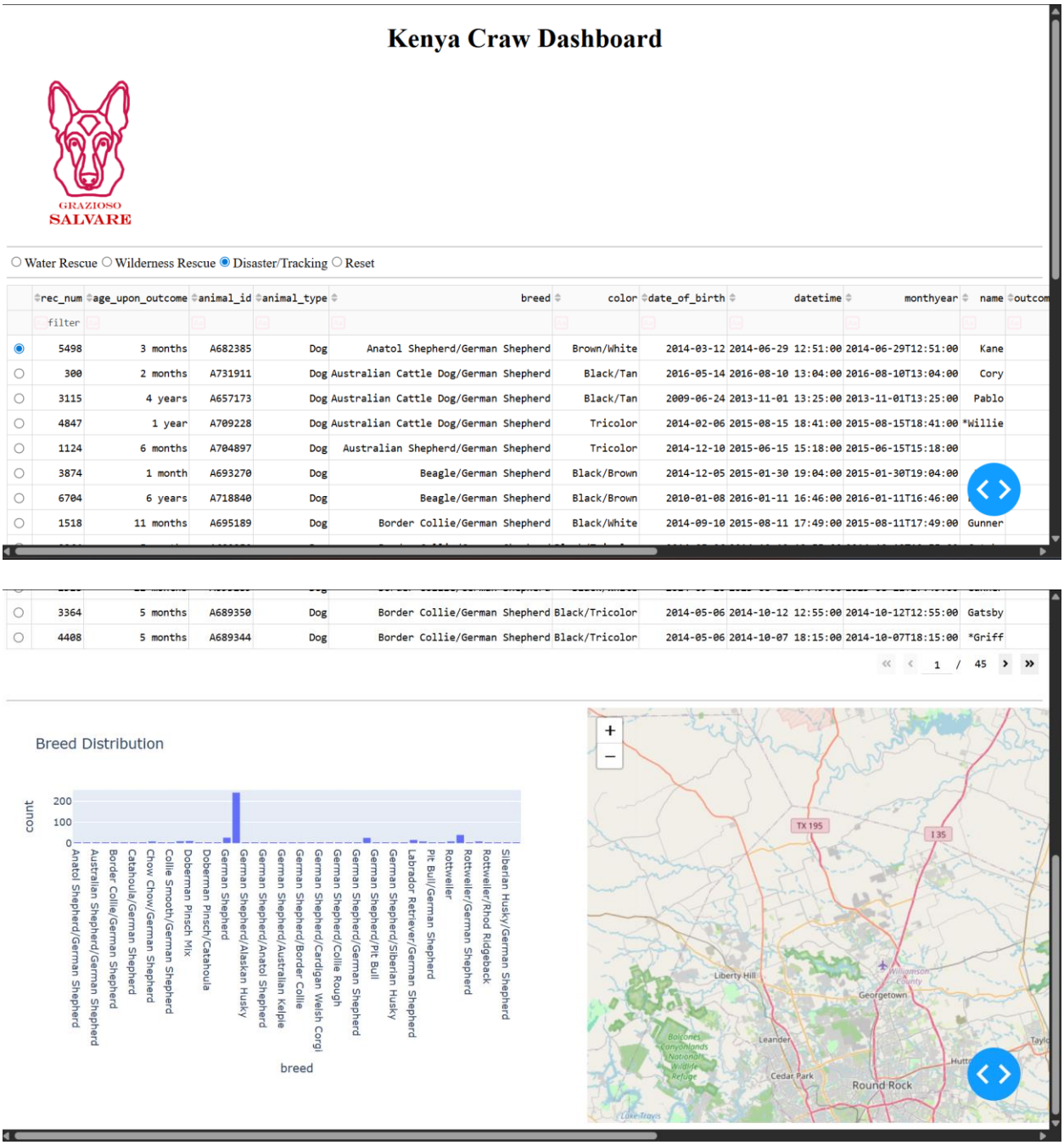
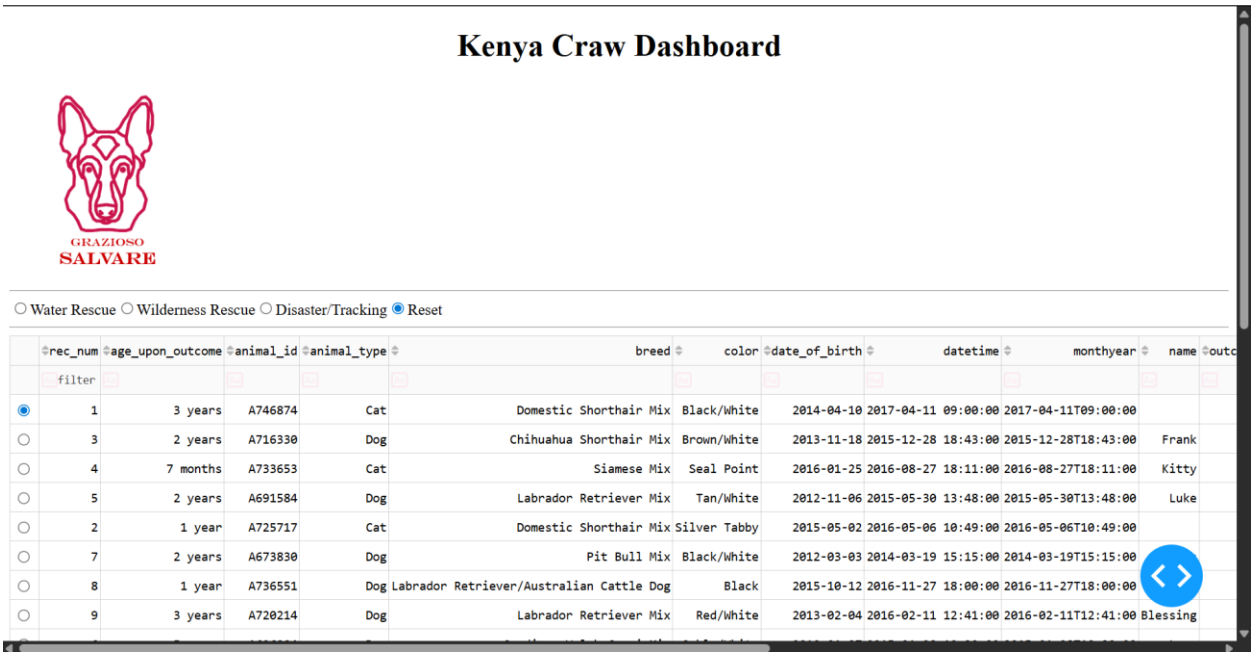


Figure 13: Reset filter (all data restored)

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().



rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outc
filter										
1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		
3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank	
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2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06T10:49:00		
7	2 years	A673830	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15:00	2014-03-19T15:15:00		
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9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing	
6	5 years	A696004	Dog	Cardigan Welsh Corgi Mix	Sable/White	2010-01-27	2015-01-28 10:39:00	2015-01-28T10:39:00	Lucy	
10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor	

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Contact

Your name: Kenya Craw

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